

Newsletter

UTHM Industrial Visit



Summary

On 3 December 2025, we visited Gerbang Digital at Universiti Tun Hussein Onn Malaysia (UTHM) and attended a series of talks delivered by the staffs from various roles. The session covered three main areas. First, we were introduced to the Pelan Strategik ICT UTHM, where the speaker explained the previous ICT plans used by the university and provided an overview of how ICT initiatives were implemented and sustained over the years. Next, we learned about the adoption of Enterprise Architecture (EA) in UTHM, which is currently recognised as an emerging technology. UTHM is now in the process of implementing EA and has prepared a draft as a reference, demonstrating its importance for institutions to learn and adapt. Finally, the visit ended with a visit to UTHM's new data center and an interactive Q&A session between the students and the UTHM staff. Through this session, we gained insights into the real-world challenges of building and operating a data centre, including managing power and water consumption, ensuring network connectivity and addressing server-related issues. We also learned about the behind-the-scenes challenges of maintaining UTHM's systems, giving us a clearer understanding of how the platforms we use, such as course registration and e-learning, are supported and maintained from the university's perspective.

Technology Used and Challenges

They are using a phased approach with the Total Caliper Information System (TCIS) to migrate from a legacy system to a new enterprise system that utilises web technology. For their server setup, they employ both hot aisle containment and cold aisle containment to optimise cooling. In terms of backup infrastructure for data recovery, they replicate their servers at the DRC in Pagoh, provided by JDT, and the PDHA provided by the government to store non-critical data, such as website information. Other security measures include the installation of firewalls, the implementation of a WAF and the establishment of a security committee to address cyberattacks.

The major challenge they faced was the lack of experience in Enterprise Architecture, which became a significant obstacle for them. However, with meticulous planning and by seeking information and help from relevant government agencies, they were able to bring their plans to execution.

They also encountered challenges in building the data center. One of the major issues was the connection to the network. From a logical perspective, the challenges include issues with network connections, IP address configuration, especially with the incorporation of IPv6, wiring, LAN connections, and other various configurations. From a physical perspective, there can be issues where the servers don't match the specifications required to fit properly in the server racks.

Reflection

Goe Jie Ying A23CS0224

I learned about the real-world challenges of running a data center. It turns out that building and operating one is not as easy as it seems. Most importantly, a data center requires a significant amount of water and electricity to function. Many data centers end up shutting down or facing operational issues due to the extremely high power and water consumption required to keep them running. I also learned how the university manages students' data and information. With multiple layers of protection in place, the data is stored securely and handled with strict safety measures. Overall, the experience gave me a clearer understanding of the complexity behind ICT operations and the importance of continuous improvement in digital infrastructure. This visit also made me realize that understanding EA is important for our future, and it is something we should learn properly to prepare ourselves for the industry.

Lam Yoke Yu A23CS0233

During this visit, instead of observing a well-established Enterprise Architecture, we visited a university that is still in the process of building one. The journey will certainly take them from "zero to hero." We came to understand their motivation for developing an EA and a data center, the challenges they faced, and how they are putting their plans into action. Although they may not yet have extensive experience, the initiative they have taken to plan the project and actively work on it is admirable. It is something we can learn from, especially since it is a large-scale project that requires significant thought and coordination.

Due to the ongoing digital transformation, universities are increasingly required to build their own enterprise systems, and this has led UTHM to begin this process. Even though they are not yet experienced, they seek support from other departments such as the Jabatan Digital Negara (JDN) and the Institut Tadbiran Awam Negara (INTAN). We also learned that maintaining a data center is not easy. The power and water consumption are major ongoing expenses, and the physical manpower required to ensure the continuous functioning of the servers is also significant.